

Shipping & tracking — research brief for Donna

Date: 2026-08-04
Audience: Donna Adams (operations) + engineering
Volume assumption: ~20–100 packages/month
Status: Research only — not implemented yet

1. Executive summary

Volunteers who order kits and gear need a **tracking number**, an **email when the package ships**, and a way to **see that status in the app**. You need the same information on the admin side for each order/user.

Recommendation: Start with a **manual hybrid** — buy discounted labels on [Pirate Ship](#) (free software; you pay postage), paste carrier + tracking into the admin order page we already have, then automatically email the volunteer via **Resend** and show tracking in Order History. When pasting labels becomes painful at volume, move to **Shippo** so labels and live tracking updates happen inside our admin console.

Do **not** rebuild the shop on Shopify for shipping, wire three separate carrier APIs (USPS/UPS/FedEx) for MVP, or add a tracking-only vendor (AfterShip) if we will buy labels through Shippo anyway. Keep **Stripe for payments only** when that ships.

2. What we need

Who	Need
Volunteer	Tracking ID when shipped; email with link; in-app “track this package”
Donna	Per-order address, carrier, tracking #, status; path to print/buy labels
Org	Fit ~20–100 pkgs/month without heavy software fees or warehouse tooling

3. What we already built

Shipping is not starting from zero.

Piece	Today
Orders database	Supabase <code>shop_orders</code> already stores <code>shipping_address</code> , <code>tracking_number</code> , <code>carrier</code> , and statuses including <code>shipped</code>
Admin	Orders list + order detail; fulfillment form lets you set status, carrier (USPS / UPS / FedEx), and tracking number
Mobile	Checkout collects a shipping address and can create an order row; Order History is not yet fully wired to live tracking
Email	Resend is live for sessions/events — no “your order shipped” email yet
Payments	Stripe not wired yet; ops shipping UX can land before or alongside payment

Gaps: buying postage, auto-email on ship, and volunteer-facing track link/status.

4. Options compared

Option	Labels?	Live track?	Fit for CUGB	Notes
Manual + Pirate Ship	Yes (in Pirate Ship website)	Tracking # + carrier website link	Best Phase 1	No public API — paste into our admin. Free tool; pay postage only.
Shippo	Yes (API + web app)	Yes (webhooks; free on labels bought via Shippo)	Best Phase 2	Donna-friendly UI <i>and</i> developer API. Free / low-cost at our volume.
EasyPost	Yes (API-first)	Yes	Strong later	Excellent API; thinner ops UI — you work mostly through <i>our</i> admin.
Native USPS / UPS / FedEx APIs	Yes	Per carrier	Poor for MVP	High setup (accounts, OAuth, separate schemas); wrong cost/effort now.
Tracking-only (AfterShip, etc.)	No	Yes (if numbers pasted elsewhere)	Usually skip	Extra vendor after postage; Shippo already covers tracking for its labels.

Option	Labels?	Live track?	Fit for CUGB	Notes
Shopify	Yes (Shopify Shipping)	Yes	Overkill	We already own Expo shop + admin + volunteers/sessions. Don't move the catalog there just for labels.

Pirate Ship vs Shippo: Pirate Ship is ideal for *buying cheap labels by hand*. It does not offer a public API. Shippo is the path when we want "Buy label" inside the Clean Up Give Back admin.

5. Recommended path

Phase 1 — Manual hybrid (about 1–2 weeks of eng)

1. Volunteer places order (address already stored on `shop_orders`).
2. Donna copies the address from admin order detail, buys a USPS/UPS label on **Pirate Ship**, packs the kit.
3. Donna enters **carrier + tracking number**, marks order **Shipped**.
4. App sends a **Resend** email: "Your order shipped" with tracking ID and a link to the carrier tracker.
5. Mobile Order History shows status, carrier, tracking #, and a **Track package** button (opens the carrier site).

No new shipping vendor account required beyond Pirate Ship (and we already use Resend).

Phase 2 — Shippo automation (when paste volume hurts)

1. Admin "Buy label" for an order: Shippo rates → pick service → purchase → store tracking (+ optional label PDF).
2. Shippo **webhooks** update shipping status (e.g. pre-transit → in transit → delivered).
3. Optional: richer in-app status text; still one branded Resend notification when first marked shipped.

Rough timeline after Phase 1 ships: add Shippo when weekly label typing becomes the bottleneck (often after a month or two at 20–100/mo).

Explicitly out of Phase 1

- Charging real shipping rates at checkout (shop still shows free shipping).
- Returns, international duties, inventory SKU system.

- Moving shop to Shopify.

6. Cost & ops notes (~20–100 packages/month)

Verify prices on vendor pages before signing up; they change.

Item	Rough cost
Pirate Ship	\$0 software; postage only
Shippo App	Free up to ~30 labels/mo; Pro roughly \$17–19/mo covers ~1–200 labels (postage extra)
Shippo API Starter	30 free labels/mo , then about 7¢ per label (API pricing)
Shippo tracking	Included for labels bought through Shippo; ~ \$0.01 per external tracking number if registered for live updates
Resend	Already configured — shipping mail is templates + triggers, not a new vendor
Postage	Main ongoing cost either way (kits/totes → typically USPS Priority / Ground Advantage–class range)

At 20–100 packages/month, Phase 1 stays cheapest. Phase 2 Shippo fees stay small relative to postage.

Ops tip: One fixed ship-from address (home/warehouse) and standard box sizes for kit vs tote make both Pirate Ship and Shippo much faster.

7. What everyone sees (target experience)

Volunteer

1. Place order → confirmation with order ID.
2. Email when Donna marks shipped (tracking ID + link).
3. Order History: status, carrier, tracking #, tap to track.

Donna

1. Orders list → open order → see who ordered what and the ship-to address.
 2. Phase 1: paste carrier + tracking after Pirate Ship.
 3. Phase 2: buy label in-admin; see live status updates.
 4. Same data available when reviewing a specific volunteer (via their shop orders).
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8. Open questions for Donna

1. **Ship-from address** — one permanent location for all kits?
 2. **Label printer** — 4×6 thermal printer, or print on paper overnight?
 3. **Preferred carriers** — USPS-first, or USPS + UPS as needed?
 4. **Who packs** — always Donna, or rotating volunteers?
 5. **When to automate** — stay on Pirate Ship until X packages/week?
 6. **Free shipping** — keep shipping free forever, or pass cost through later with Stripe?
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9. Sources

- Pirate Ship — [Does Pirate Ship have an API?](#) (no public API)
 - Shippo — [API pricing](#), [App pricing](#), [Tracking](#), [Webhooks](#)
 - Internal ground truth — `shop_orders` schema (`admin/db/001_admin_portal_migration.sql`),
admin order fulfillment UI (`admin-web-app`), Resend status
(`docs/backend/context/payments.md`)
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Bottom line for Donna

Start simple: Pirate Ship for labels, our existing Orders screen for tracking entry, Resend for email, app Order History for volunteers. **Graduate to Shippo** when you're ready to buy labels and get live package updates without leaving Clean Up Give Back's admin. That path matches how we already store orders and keeps software cost low at 20–100 shipments a month.